**NAME:** ______________________ **PERIOD:** ________ **DATE:** ____________

# A Test of Mean Resolution Time

AP Statistics · Unit 4 · Topic 4.5 · B: 2027-02-02 · E: 2027-02-05

## [STAR] TODAY'S PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

**Leah:** "The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide."

**Damon:** "Because $p > 0.05$, accept $H_0$. The test proves the population mean was exactly 12 hours and the observed difference was only chance."

**Ticket sample**

| Summary | Value |
|---|---|
| **Population** | 1200 |
| $n$ | 24 |
| **Mean (h)** | 11.1 |
| **SD (h)** | 2.8 |
| **Shape** | Symmetric; no outliers |

### FIRST TAKE — before we discuss (5 min)

Does this small study seem able to prove that the average was exactly 12 hours? Explain in one sentence.

### [BOOK] COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)

For a one-sample mean test with unknown population standard deviation, $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ has $df = n - 1$. The $p$-value is the probability, assuming $H_0$ is true, of a test statistic at least as extreme as the observed statistic in the direction of $H_a$. If $p \le \alpha$, reject $H_0$; if $p > \alpha$, fail to reject $H_0$. Failing to reject says the sample does not provide convincing evidence for $H_a$ at that significance level. It does not prove the null value, and the conclusion must concern the population represented by the random sample.

**Work (a)–(c) from the rules box:**

DOK 1 **(a)** State $H_0$ and $H_a$ for the population mean resolution time.

DOK 2 **(b)** Using $\bar{x} = 11.1$, $s = 2.8$, and $n = 24$, calculate $t$. Use the supplied $p = 0.0645$; you do not need to find it again.

DOK 3 **(c)** In 3–4 sentences, decide whose account is justified. Compare the p-value with 0.05, state the conclusion for all 1,200 tickets, and explain why failure to reject does not prove equality or that chance was the only explanation.

**Frame:** The numerical comparison is $p$ ________ $\alpha$, so the formal decision is ____________________ and the population conclusion is ____________________________.

**Frame:** That decision __________ establish ________________________ because the p-value is calculated assuming ________________________.

**EXIT — turn this in**

One thing the rules box changed about my first take: ________________________________________